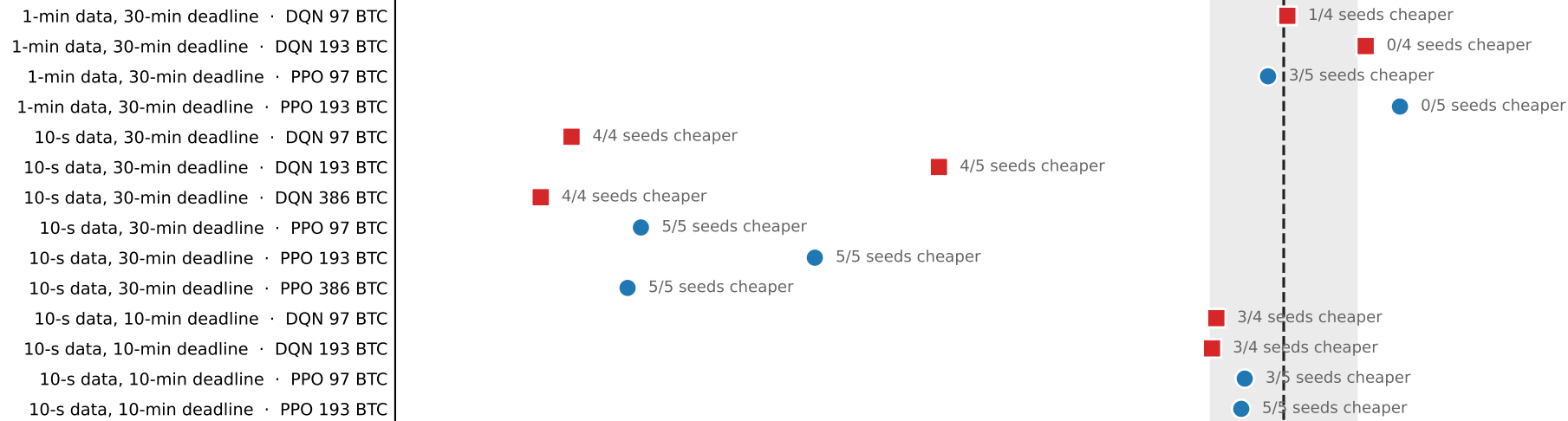


Three environments, three sealed verdicts: no stable improvement on TWAP anywhere -- including where a simple rule proves the edge exists

1. Frozen replay of real order-book data

no stable edge — 4 of 14 arms clear the bar, and a diagnosed-broken learner clears it too



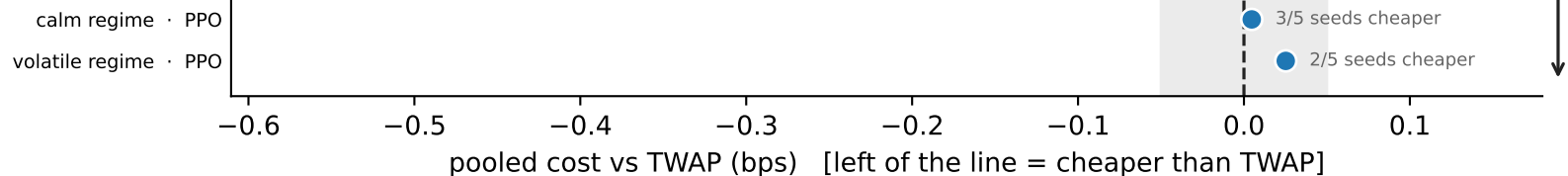
2. Reactive simulator, no injected signal

null — both sealed confirmations fail

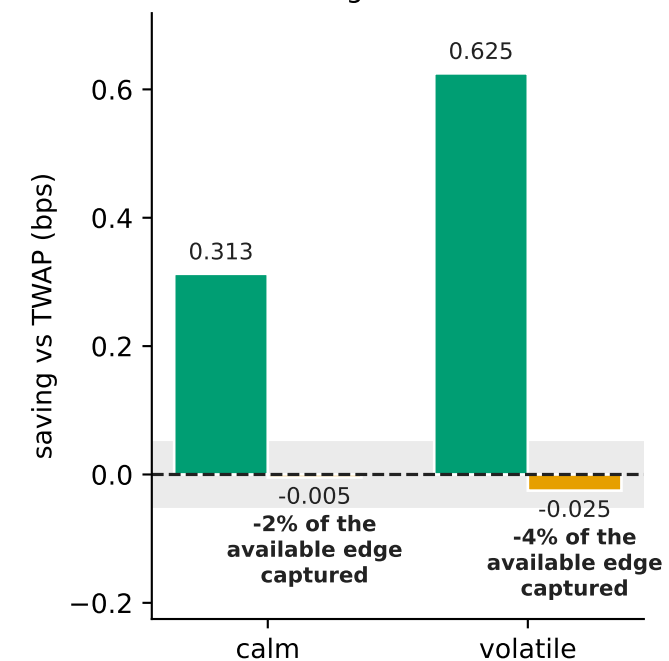


3. Reactive simulator + measured signal

null — both regimes fail



Environment 3: the edge was real and the agents left all of it



● PPO arm (sealed) ---- TWAP ■ registered rule, zero fitted parameters
 ■ DQN arm (sealed) +/-0.05 bps materiality band ■ trained agents (sealed)